Analysis of Smooth Contacting Cylinders in Equilibrium

1. High-Level Overview

In structural and mechanical engineering, determining how objects support each other and interact with constraining walls is a foundational task. When smooth cylindrical objects are placed together inside a channel or container, they exert contact forces on one another and on the boundaries.

To determine the unknown support reactions:

- **Smooth Surfaces:** A "smooth" surface implies zero friction. Therefore, reaction forces act strictly **perpendicular (normal)** to the tangent at the point of contact.
- **Line of Action:** For circular cylinders, any normal contact force acts along a line passing through the center of the circle.
- **Equilibrium:** An object at rest satisfies static equilibrium, meaning the net forces acting on it in any direction sum to zero ($\sum F_x = 0, \sum F_y = 0$). Alternatively, for three coplanar, concurrent, non-parallel forces, **Lami's Theorem** can be applied.

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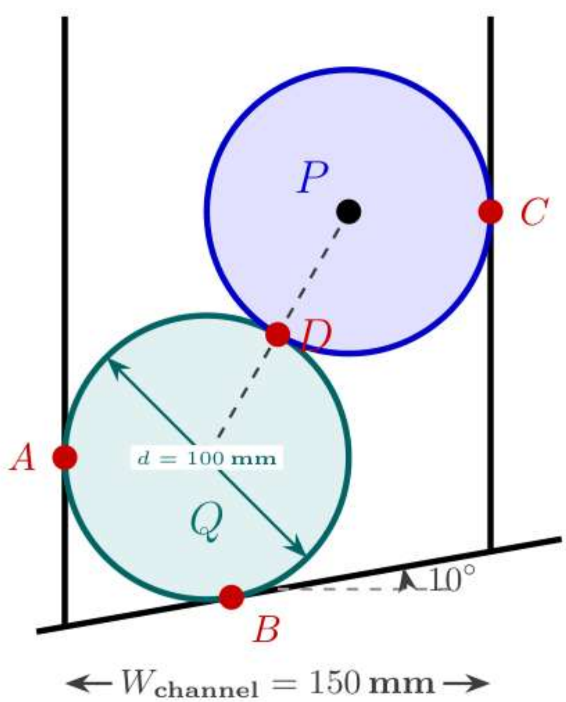
2. Fundamental Principles & Geometry Setup

Consider two identical smooth cylinders placed inside a channel.

System Parameters

- **Diameter of each cylinder (d):** $100 \text{ mm} \implies \text{Radius } (R) = 50 \text{ mm}$
- **Weight of each cylinder (W):** 200 N
- **Channel Width (W_{channel}):** 150 mm
- **Incline Angle of Base Surface:** 10° relative to the horizontal



Description of Visual Layout:
A vertical channel of width 150 mm with a bottom floor inclined upwards to the right at an angle of 10°. Two cylinders are stacked inside: lower

cylinder Q contacts the left vertical wall at A and the inclined floor at B. Upper cylinder P rests on top of cylinder Q at contact point D and touches the right vertical wall at C. Center of lower cylinder is Q, center of upper cylinder is P.

Contact Reaction Summary

1. **At Point A (Left Vertical Wall on Cylinder Q):** Reaction force R_A acts horizontally to the right.
2. **At Point B (Inclined Floor on Cylinder Q):** Reaction force R_B acts perpendicular to the 10° inclined surface, directing inward toward center Q.
3. **At Point C (Right Vertical Wall on Cylinder P):** Reaction force R_C acts horizontally to the left.
4. **At Point D (Contact between Cylinder P and Q):** Mutual normal contact reaction R_D acts along the line connecting centers Q and P.

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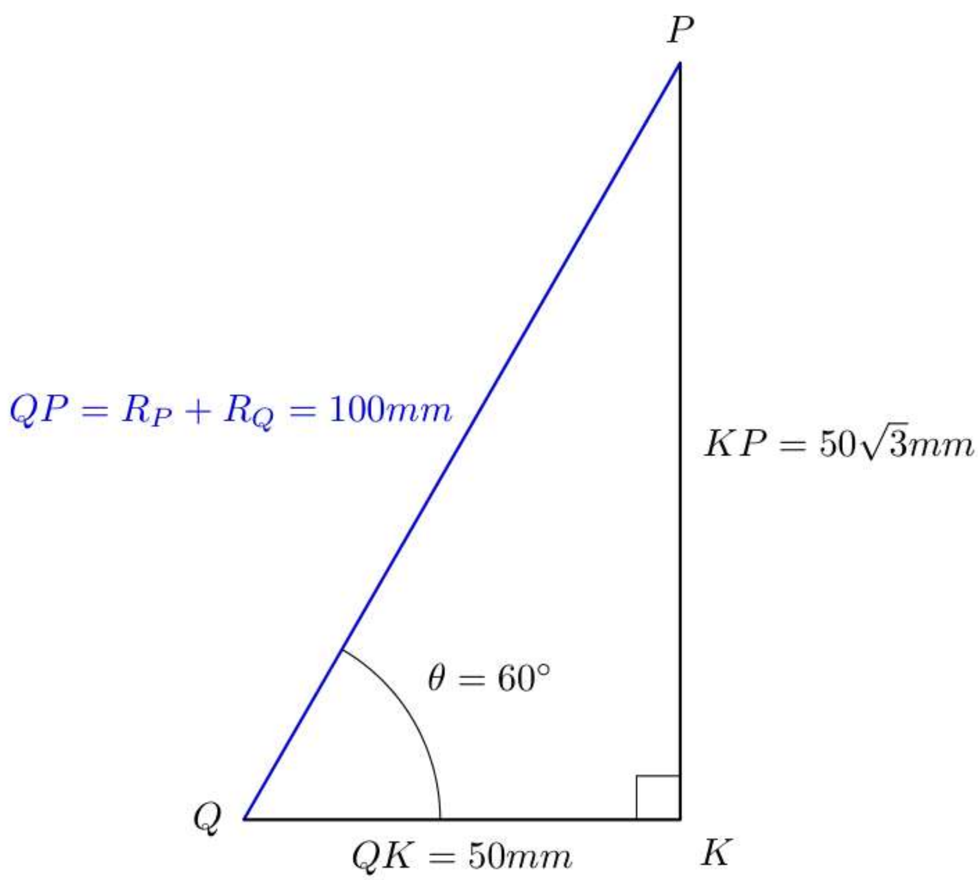
3. Geometric Derivations

To analyze force components, we must determine the angle of inclination of the line connecting centers Q and P, denoted as angle θ.



Description of Visual Layout:
A right-angled triangle formed by hypotenuse QP, horizontal leg QK, and vertical leg KP. Point Q is the center of the left cylinder, P is the center of the right cylinder, and K is a point horizontally level with Q and vertically aligned with P. Angle $\angle PQK = \theta$.

Step 1: Calculate the Hypotenuse (QP)

The distance between centers Q and P equals the sum of their radii:

$$QP = R + R = 50 \text{ mm} + 50 \text{ mm} = 100 \text{ mm}$$

Step 2: Calculate the Horizontal Distance (QK)

The total channel width is 150 mm.

- Center Q is located at distance $R = 50 \text{ mm}$ from the left wall.
- Center P is located at distance $R = 50 \text{ mm}$ from the right wall.

Therefore, the horizontal distance between centers Q and P is:

$$QK = W_{\text{channel}} - (R + R) = 150 \text{ mm} - (50 \text{ mm} + 50 \text{ mm}) = 50 \text{ mm}$$

Step 3: Compute Angle θ

Using the cosine relation in right triangle $\triangle QKP$:

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{QK}{QP}$$

$$\cos(\theta) = \frac{50}{100} = 0.5$$

$$\theta = \arccos(0.5) = 60^\circ$$

Thus, the line connecting the centers of the two cylinders is inclined at 60° **to the horizontal**.

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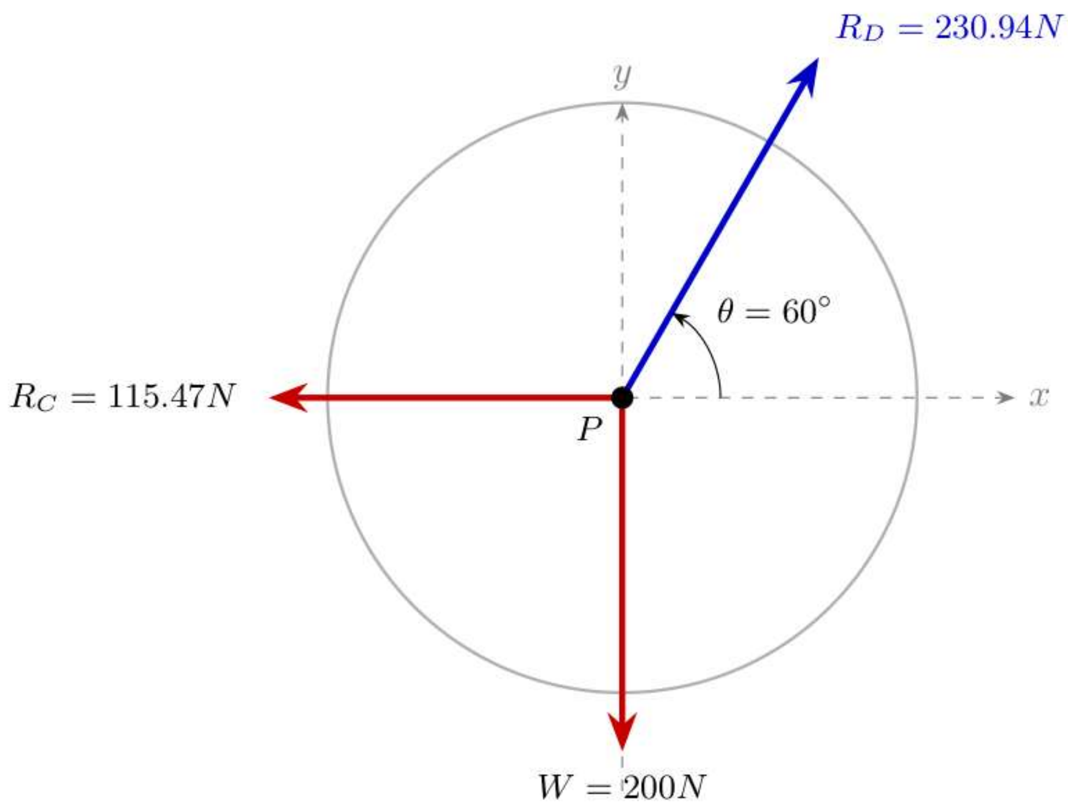
4. Equilibrium Analysis of Cylinder P (Upper Cylinder)

Cylinder P is acted upon by three concurrent forces:

1. Self-weight ($W = 200 \text{ N}$) acting vertically downwards.
2. Reaction from right wall (R_C) acting horizontally to the left.
3. Reaction from lower cylinder (R_D) acting along the line of centers at an angle of 60° above the horizontal, directed towards center P.



Description of Visual Layout:
A point P showing three force vectors: downward weight $W = 200$ N, horizontal force R_C pointing left, and pushing force R_D acting along a line 60° relative to the horizontal pointing up and to the right.

Application of Lami's Theorem

Lami's Theorem states that for three coplanar, concurrent forces in equilibrium:

$$\frac{F_1}{\sin(\alpha)} = \frac{F_2}{\sin(\beta)} = \frac{F_3}{\sin(\gamma)}$$

where α, β, γ are the angles opposite to forces F_1, F_2, F_3.

Determining Angles Between Forces at Center P:

1. Angle between W (200 N) and R_C: 90°
2. Angle between R_C and R_D:
   - R_C is pointing along the negative x-axis (180°).
   - R_D line extends at 60° to the horizontal.
   - Angle between R_C and $R_D = 180^\circ - 30^\circ = 150^\circ$
3. Angle between W (200 N) and R_D: $90^\circ + 30^\circ = 120^\circ$

Formulating Equations:

$$\frac{200}{\sin(120^\circ)} = \frac{R_C}{\sin(150^\circ)} = \frac{R_D}{\sin(90^\circ)}$$

Solving for R_C:

$$R_C = 200 \cdot \frac{\sin(150^\circ)}{\sin(120^\circ)} = 200 \cdot \frac{0.5}{\frac{\sqrt{3}}{2}} = \frac{200}{\sqrt{3}} \approx 115.47\ \mathrm{N}$$

Solving for R_D:

$$R_D = 200 \cdot \frac{\sin(90^\circ)}{\sin(120^\circ)} = 200 \cdot \frac{1}{\frac{\sqrt{3}}{2}} = \frac{400}{\sqrt{3}} \approx 230.94\ \mathrm{N}$$

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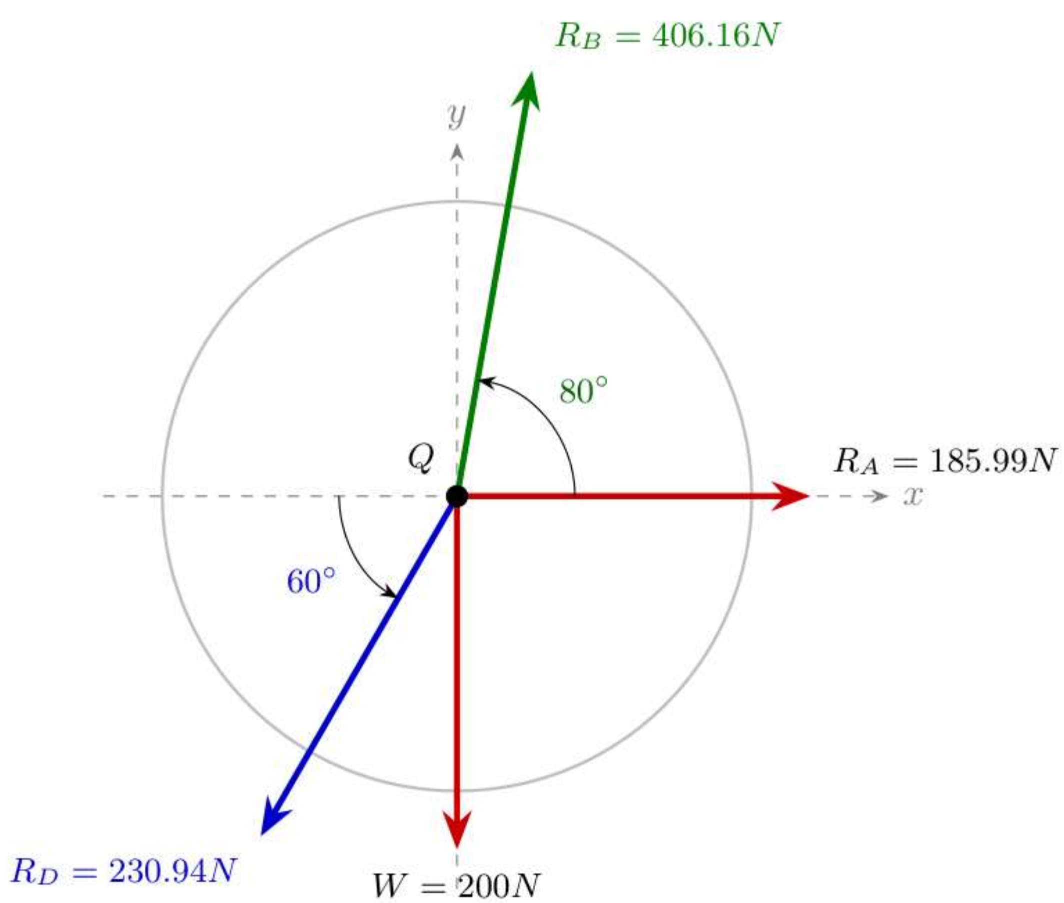
5. Equilibrium Analysis of Cylinder Q (Lower Cylinder)

Cylinder Q is subjected to four forces:

1. Self-weight ($W = 200\ \mathrm{N}$) acting vertically downward.
2. Reaction from left vertical wall (R_A) acting horizontally to the right.
3. Reaction from upper cylinder ($R_D = 230.94\ \mathrm{N}$) pushing downwards along the line of centers at 60° to the horizontal (by Newton's Third Law).
4. Reaction from the inclined floor (R_B).



Description of Visual Layout:
Origin Q showing vector R_A along positive x-axis, downward force $200\ \mathrm{N}$, contact force R_D pointing down-left at 60° to horizontal, and reaction R_B pointing up-right perpendicular to the 10° base inclined floor.

Step 1: Determining Angle of Reaction R_B

- Floor inclination = 10° with horizontal.
- Since R_B is perpendicular to the inclined plane, its angle with the vertical is 10°, which corresponds to an angle of $90^\circ - 10^\circ = 80^\circ$ with the horizontal.

Step 2: Resolving Force Components

Forces along Vertical axis ($\sum F_y = 0$):

$$\sum F_y = -W - R_D \sin(60^\circ) + R_B \sin(80^\circ) = 0$$

Substitute known values ($W = 200 \text{ N}$, $R_D = 230.94 \text{ N}$):

$$-200 - (230.94)\sin(60^\circ) + R_B \sin(80^\circ) = 0$$

$$-200 - 200 + R_B \sin(80^\circ) = 0$$

$$-400 + R_B(0.9848) = 0$$

$$R_B = \frac{400}{0.9848} \approx 406.16 \text{ N}$$

Forces along Horizontal axis ($\sum F_x = 0$):

$$\sum F_x = R_A - R_D \cos(60^\circ) - R_B \cos(80^\circ) = 0$$

Substitute known values ($R_D = 230.94 \text{ N}$, $R_B = 406.16 \text{ N}$):

$$R_A - (230.94)\cos(60^\circ) - (406.16)\cos(80^\circ) = 0$$

$$R_A - (230.94 \times 0.5) - (406.16 \times 0.1736) = 0$$

$$R_A - 115.47 - 70.52 = 0$$

$$R_A = 185.99 \text{ N}$$

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6. Summary of Results

Support Reaction	**Location**	**Magnitude**	**Direction**
R_A	Left Wall (at A)	185.99 N	Horizontally to the right ($\rightarrow$)
R_B	Inclined Base (at B)	406.16 N	Inclined at 80° to horizontal ($\nearrow$)
R_C	Right Wall (at C)	115.47 N	Horizontally to the left ($\leftarrow$)
R_D	Contact Point (at D)	230.94 N	Along line of centers at 60°